

BULLHORN HEADWATERS, AB, CanadaWMO: **710160**Lat: **49.153N**Lon: **113.551W**Elev: **4350**StdP: **12.53**Time Zone: **-7.00 (NAM)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.0	-10.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.2	83.7	N/A	80.4	N/A	77.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	24.0	22.0	30.8	37.5	47.8	54.8	61.9	60.8	52.3	40.6	31.9	22.3		
	DBStd	17.61	14.25	15.09	10.79	9.69	7.93	6.11	5.77	6.32	8.69	10.05	12.60	13.81		
	HDD50	4528	807	784	594	383	139	21	1	4	73	313	549	860		
	HDD65	8950	1271	1204	1059	824	534	311	129	157	387	755	994	1325		
	CDD50	1124	0	0	0	9	71	164	370	340	142	23	5	0		
	CDD65	71	0	0	0	0	1	4	33	28	5	0	0	0		
	CDH74	1444	0	0	0	1	8	77	622	585	150	1	0	0		
	CDH80	327	0	0	0	0	0	11	164	134	18	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	28.7	1.9	1.6	2.0	2.6	3.6	4.7	2.0	2.1	2.3	2.0	2.0	1.8		
	PrecMax	42.9	4.3	3.0	3.0	5.1	8.4	12.7	7.2	5.7	5.7	4.8	3.8	3.5		
	PrecMin	20.5	0.6	0.4	0.7	1.1	0.5	0.8	0.2	0.4	0.2	0.5	0.7	0.9		
	PrecStd	5.7	0.8	0.6	0.7	0.8	1.9	2.9	1.7	1.4	1.3	1.0	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.8	52.3	55.4	70.3	75.2	81.6	89.2	87.3	82.6	69.2	63.3	49.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	46.0	47.2	51.0	62.7	71.5	75.5	84.5	83.5	78.3	64.0	55.8	45.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	43.5	43.2	48.0	57.2	68.1	71.8	80.5	80.5	74.4	59.1	51.0	42.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	41.0	39.8	44.8	51.7	63.3	68.3	77.0	77.0	69.2	54.9	47.3	38.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.2	16.2	16.8	17.3	18.8	19.9	24.2	25.4	22.3	17.5	15.4	14.7		
		MCDBR	16.4	16.6	19.8	25.8	27.0	26.7	31.1	32.5	32.5	23.0	18.9	16.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.237	0.249	0.272	0.324	0.331	0.325	0.326	0.341	0.308	0.276	0.251	0.226		
	taud		2.401	2.393	2.416	2.318	2.406	2.470	2.476	2.422	2.490	2.561	2.505	2.414		
	Ebn at Noon		271	293	302	293	295	296	294	285	285	280	264	261		
	Edh at Noon		20	25	30	37	36	34	33	33	28	22	18	17		
(44) All-Sky Solar Radiation	RadAvg		390	707	1093	1511	1801	2010	2252	1805	1282	789	435	312		
	RadStd		32	53	98	100	158	158	129	116	107	77	33	29		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page